Tabla de Multiplicación

miércoles, 29 de octubre de 2025 03:57 p. m.

Binario	Rep Polinomial
0 0 0	0
0 0 1	1
0 1 0	x
0 1 1	$x+1$
1 0 0	x^2
1 0 1	x^2+1
1 1 0	x^2+x
1 1 1	x^2+x+1

*	1	x	$x+1$	x^2	x^2+1	x^2+x	x^2+x+1
1	1	x	$x+1$	x^2	x^2+1	x^2+x	x^2+x+1
x	x	x^2	x^2+x	$x+1$	1	x^2+x+1	x^2+1
$x+1$	$x+1$	x^2+x	x^2+1	x^2+x+1	x^2	1	x
x^2	x^2	$x+1$	x^2+x+1	x^2+x	x	x^2+1	1
x^2+1	x^2+1	1	x^2	x	x^2+x+1	$x+1$	x^2+x
x^2+x	x^2+x	x^2+x+1	1	x^2+1	$x+1$	x	x^2
x^2+x+1	x^2+x+1	x^2+x	x	1	x^2+x	x^2	$x+1$

$x \cdot x^2 \bmod (x^3+x+1) = x^3 \bmod (x^3+x+1) = x+1$

$x(x+1) \bmod (x^3+x+1) = x^2+x \bmod (x^3+x+1) = x^2+x$

$x^2 \cdot (x^2+1) \bmod (x^3+x+1) = x^4+x^2 \bmod (x^3+x+1) = x^4 \bmod (x^3+x+1) + x^2 \bmod (x^3+x+1)$

$x \cdot x^3 \bmod (x^3+x+1) + x^2$

$x \cdot x+1 + x^2$

$x^2+x+x^2 = x$

$(x^2+x+1)(x+1) \bmod (x^3+x+1) = x^3+x^2+x^2+x+x+1 \bmod (x^3+x+1)$

$= x^3+1 \bmod (x^3+x+1) = x$

$x(x^2+x) \bmod (x^3+x+1) = x^3+x \bmod (x^3+x+1) = x+1+x^2 = x^2+x+1$

$(x^2+x+1)(x) \bmod (x^3+x+1) = x^3+x^2+x \bmod (x^3+x+1) = m^2+1$

$(x+1)(x+1) \bmod (x^3+x+1) = x^2+2x+1 \bmod (x^3+x+1) = x^2+1$

$x^2(x+1) \bmod (x^3+x+1) = x^3+x^2 \bmod (x^3+x+1) = x^2+x+1$

$(x^2+1)(x+1) \bmod (x^3+x+1) = x^3+x^2+x+1 \bmod (x^3+x+1) = x^2$

$(x^2+x)(x+1) \bmod (x^3+x+1) = x^3+x^2+x^2+x \bmod (x^3+x+1) = x^3+x \bmod (x^3+x+1) = 1$

$(x^2)(x^2) \bmod (x^3+x+1) = x^4 \bmod (x^3+x+1) = x \cdot x^3 \bmod (x^3+x+1) = x \cdot x+1 = x^2+x$

$(x^2)(x^2+1) \bmod (x^3+x+1) = x^4+x^3 \bmod (x^3+x+1) = (x^2+x)+(x+1) = x^2+1$

$(x^2)(x^2+x+1) \bmod (x^3+x+1) = x^4+x^3+x^2 \bmod (x^3+x+1) = (x^2+x)+(x+1)+x^2 = 2x^2+2x+1 = 1$

$(x^2+1)(x^2+1) \bmod (x^3+x+1) = x^4+x^2+x^2+1 \bmod (x^3+x+1) = x^2+x+1$

$(x^2+1)(x^2+x) \bmod (x^3+x+1) = x^4+x^3+x^2+x \bmod (x^3+x+1) = (x^2+x)+(x+1)+x^2+x = 2x^2+x+1 = x+1$

$(x^2+1)(x^2+x+1) \bmod (x^3+x+1) = x^4+x^3+x^2+x^2+x+1 \bmod (x^3+x+1) = (x^2+x)+(x+1)+x+1 = x^2+1$

$(x^2+x)(x^2+x) \bmod (x^3+x+1) = x^4+x^3+x^3+x^2 \bmod (x^3+x+1) = x^4+x^2 \bmod (x^3+x+1) = (x^2+x)+x^2 = x$